

# llmXive Automated Review of Dockerless: Environment-Free Program Verifier for Coding Agents

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** [qwen.qwen3.5-122b](#)

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents quantitative claims that are largely consistent with the provided tables, but three specific areas require clarification to ensure the text accurately reflects the underlying evidence.

First, in Section 3.2, the description of the inference mechanism for the continuous score  $r_\phi(x, y)$  contains a potential ambiguity. The text states the model outputs a “binary token” and derives the score from the “logits of the two verdict tokens.” The provided formula is a standard softmax over two logits. While mathematically consistent, the phrasing could be misinterpreted as referring to the logits of two *generated* tokens in a sequence rather than the logits of the two possible next tokens (0 or 1) at a single position. Given the training objective is next-token prediction, the claim should explicitly confirm that the score represents the probability of the “1” token at the final judgment step.

Second, regarding the dataset scale in Section 3.2 and the Abstract, the paper claims training on “3.7K issues.” Appendix A.1 clarifies this covers “3.7K unique issues” after cleaning. However, the main text describes the process as “rejection sampling on 3.7K issues... retaining only... trajectories.” This phrasing risks implying that 3.7K is the size of the *final* training set. Since rejection sampling discards trajectories where the predicted verdict does not match the ground truth, the final number of training examples is strictly less than the source count. The claim should specify that 3.7K is the number of *source* issues to avoid overstating the volume of labeled training data.

Finally, the performance claims in Section 4.1 are numerically accurate (62.0% vs 59.6% for Qwen, and 62.0% vs 41.2% for SWE-Lego). However, the text groups these comparisons closely. To ensure the claim “surpasses the Qwen3.5-9B baseline” is not conflated with the comparison to “the next-best open-source SWE specialist,” the text should maintain a clear distinction between the *base model* improvement and the *state-of-the-art* specialist improvement, as the magnitude and significance of these two claims differ.

**Action items.**

- **[writing]** Clarify in Section 3.2 if the score  $r_\phi$  is derived from logits of generated tokens or a classification head to align the ‘binary token’ claim with the softmax formula.
- **[writing]** In Section 4.1, clearly distinguish between the Qwen3.5-9B baseline improvement (2.4 pts) and the SWE-Lego-8B specialist improvement (20.8 pts) to avoid conflating the two claims.
- **[writing]** In Section 3.2, specify if ‘3.7K issues’ refers to the source pool or the final training set size after rejection sampling to avoid overstating the labeled data volume.

**Figure Critic**

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

The figure effectively visualizes the comparison between Docker-based tests, LLM scorers, and the proposed Dockerless method. However, the caption contains a grammatical error where the subject is missing from the final sentence, and the description of the LLM scorer could be more precise regarding its lack of repository access.

### Figure 2

The figure provides a clear and well-structured visual architecture of the system. However, the caption contains a missing model name and a minor LaTeX formatting error in the score notation.

### Figure 3

Figure 3 presents resolved rates with solid and hatched bars, but the legend label ‘w/o Env’ for the hatched portion contradicts the caption’s explanation that the hatched part shows the gain from per-repository environments. This creates ambiguity about whether the hatched bar represents ‘without environment’ or the ‘environmental gain’.

### Figure 4

The figure clearly illustrates the training pipeline with distinct stages and logical flow. However, the caption is missing the model name, and the diagram contains a spelling error (‘Pathes’ instead of ‘Paths’).

### Figure 5

The figure clearly illustrates the two-stage training pipeline (RFT and RL) with distinct visual blocks for each step. However, the caption text is defective, omitting the model name ‘Dockerless’ in several key phrases, which obscures the description of the pipeline components.

### Figure 6

The figure effectively demonstrates the performance trend of the verifier against the number of questions, but the non-uniform x-axis scaling and the ambiguous boundary of the ‘sweet spot’ region reduce graphical precision.

## Figure 7

The figure effectively highlights the cost breakdown but fails to label the three reward sources corresponding to the bars, and the visual bar lengths do not accurately reflect the sum of the shared rollout cost and the added reward evaluation time described in the annotations.

## Figure 8

Figure 8 effectively illustrates the candidate vs. reference patch and the verifier’s evidence collection, but the interpretation of the ‘Similarity’ score and the lack of explicit definition for the ‘Dockerless’ score as a pass/fail metric reduce clarity.

## Figure 9

The figure effectively visualizes performance differences, but the legend label ‘w/o Env’ for the hatched bars contradicts the caption’s explanation that these bars represent the gain from per-repository environments, creating significant confusion regarding the data representation.

## Figure 10

The figure effectively visualizes the per-language comparison with clear diagonal separation, but it lacks a legend for bubble sizes and explicit percentage units on the axes.

## Figure 11

The figure fails to visualize the comparison described in the caption; while the legend lists three reward sources, only one distribution is plotted, making it impossible to verify the claim that they produce ‘near-identical distributions’.

**Action items.**

- **[writing]** Figure 1 caption: The sentence ‘instead deeply explores the codebase...’ is missing the subject (the method name ‘Dockerless’), making it grammatically incomplete and unclear.
- **[writing]** Figure 1 caption: The phrase ‘LLM scorers sidestep that cost’ is ambiguous; the diagram shows the LLM Scorer lacks a repository (‘No Repo’), but the text should explicitly state it lacks repository grounding to match the visual.
- **[writing]** Figure 2 caption: The phrase ‘Architecture of ’ is missing the model name (likely ‘Dockerless’), rendering the sentence grammatically incomplete.
- **[writing]** Figure 2 caption: The score notation ‘ $r_{(x, y)}$ ’ uses incorrect LaTeX syntax; it should likely be formatted as  $r_{\phi}(x, y)$  to match the visual label in the figure.
- **[science]** Figure 3: The caption states that ‘hatched extensions (w/o Env) show the additional gain from per-repository environments,’ but the legend labels the hatched pattern as ‘w/o Env’ (without environment). This contradicts the caption’s claim that the hatched portion represents the gain from *using* environments (implying the full bar is ‘with env’). The label ‘w/o Env’ on the hatched segment suggests it represents the ‘without environment’ condition,

which conflicts with the caption’s explanation t

- **[writing]** Figure 3: The caption contains a typo: ‘hatched extensions (w/o Env) show the additional gain from per-repository environments’ is confusingly phrased. It should clarify that the hatched portion represents the *difference* between env-based and env-free scores, but the label ‘w/o Env’ on the hatched bar is misleading given the caption’s description.
- **[writing]** Figure 4: The label ‘Candidate Pathes’ contains a spelling error; it should be ‘Candidate Paths’.
- **[writing]** Figure 4: The caption contains a missing model name (‘Training pipeline for :’), likely due to a formatting error where the model name was omitted.
- **[writing]** Figure 5: The caption contains multiple missing nouns where the model name ‘Dockerless’ should appear (e.g., ‘pipeline for .’, ‘scored by .’, ‘uses as the per-rollout reward source’). The diagram itself labels the verifier as ‘Dockerless’, but the caption text is broken.
- **[writing]** Figure 6: The x-axis has non-uniform spacing between ticks (0, 1, 2, 4, 6, 8) which visually distorts the slope of the line segments, particularly between 2 and 4.
- **[writing]** Figure 6: The shaded region labeled ‘sweet spot’ spans  $x=2$  to  $x=5$ , but the x-axis has no tick mark at 5, making the right boundary of the region ambiguous.
- **[writing]** Figure 7: The caption mentions ‘three reward sources’ but the chart only displays three bars without a legend or labels identifying which source corresponds to which bar (Dockerless, Test Execution, DeepSWE Verifier).
- **[science]** Figure 7: The x-axis scale (0 to 2500s) and bar lengths do not visually align with the provided numerical annotations (e.g.,  $2308s + 180s = 2488s$ , yet the teal bar ends near 2300s), suggesting the bars represent only the variable cost or the scale is misleading.
- **[writing]** Figure 8: The caption states the candidate patch resolves the issue, but the ‘Dockerless’ score (0.996) is not explicitly defined as a ‘resolved’ or ‘pass’ metric in the figure or caption, unlike the ‘Execution Pass (1.0)’ label on the Ground Truth bar.
- **[science]** Figure 8: The ‘Similarity’ bar (0.468) is shown as a result of the ‘Similarity Check’ on the candidate patch, but the figure does not clarify whether this low similarity is expected or problematic given the caption’s claim that the candidate patch is valid.
- **[science]** Figure 9: The caption states ‘Solid bars are env-free; hatched extensions show the additional gain from per-repository environments,’ but the legend labels the hatched portion as ‘w/o Env’ (without environment). This directly contradicts the caption’s definition that the hatched part represents the gain from using environments.
- **[science]** Figure 9: The caption claims ‘full bar height equals the env-based score,’ yet the hatched bars are visually shorter than the solid bars in several cases (e.g., DS-V3.2 Pro), implying the ‘gain’ is negative or the visualization logic is inverted relative to the text description.

- **[writing]** Figure 10: The bubble size legend is missing; the caption states ‘Bubble size encodes the number of test instances per language,’ but no scale or reference bubbles are provided to interpret the values.
- **[writing]** Figure 10: Axis labels lack units; the caption mentions ‘resolve rate (%)’, but the axes only show numbers (20–70) without a ‘%’ symbol or explicit unit label.
- **[science]** Figure 11: The caption claims to show distributions for ‘three reward sources’, but the plot contains only a single filled area (likely the ‘DeepSWE Verifier’ from the legend) and no other curves or histograms to compare against.
- **[writing]** Figure 11: The legend lists ‘DeepSWE Verifier’, ‘Test Execution’, and ‘Dockerless’, but the plot only displays the data for ‘DeepSWE Verifier’, leaving the other two sources unrepresented.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The manuscript relies heavily on domain-specific acronyms and informal shorthand that may alienate non-specialist readers or those new to the specific sub-field of software engineering agents. While the paper is technically sound, the density of undefined or loosely defined jargon creates unnecessary friction.

First, the Abstract introduces “SFT” and “RL” immediately. While standard in the field, a generalist reader might not instantly map these to “supervised fine-tuning” and “reinforcement learning” without the explicit expansion provided in the very first sentence. The text does expand them, but the flow is dense. More critically, the term “AUC” appears in the Abstract (“14.3 AUC points”) without definition. While Area Under the Curve is standard, the paper’s claim to “environment-free” scalability suggests a broader audience; defining AUC briefly upon first use would improve accessibility.

Second, the paper introduces “env-free” and “env-based” as acronyms in Section 2.1. The use of “env” as a prefix is informal jargon (short for “environment”) that, while common in engineering, is not standard academic prose. The text defines them as “environment-free (env-free)”, but the subsequent heavy reliance on the hyphenated acronyms throughout the text (e.g., “env-free rollouts,” “env-free setting”) creates a barrier. It would be more professional to use the full terms or a more formal abbreviation like “EF” if space is a constraint, though the full terms are preferable for clarity.

Third, “RFT” is introduced in Section 2.3 as “Rejection-sampling fine-tuning (RFT)”. However, the acronym is used frequently in the Results section without reiteration. Given that “RFT” is not as universally recognized as “SFT” or “RL” in the broader AI community, ensuring the full term is used at least once in the Abstract or Introduction would be beneficial.

Finally, “GRPO” is used in Section 2.3 (“optimize the model with GRPO”) without expansion. The paper cites the source (Shao et al.), but the acronym itself is not defined in the text. A reader unfamiliar with the specific “Group Relative Policy Optimization” variant would be left guessing. Defining this acronym at first use is essential for a self-contained paper.

Overall, the paper is well-structured, but a pass to expand these acronyms and replace informal “env-“ shorthand with full terms would significantly improve readability for a wider audience.

#### Action items.

- **[writing]** Define ‘AUC’ at first use in the Abstract and Section 2.1. While standard in ML, the paper targets a broader audience including those unfamiliar with ROC curves. Replace ‘14.3 AUC points’ with ‘14.3 points in Area Under the Curve (AUC)’ on first occurrence.
- **[writing]** Replace the acronym ‘RFT’ with ‘rejection-sampling fine-tuning’ or ‘rejection fine-tuning’ at its first appearance in Section 2.3. The text currently introduces ‘Rejection-sampling fine-tuning (RFT)’ but later uses ‘RFT’ without ensuring the reader has retained the definition, and the acronym is not defined in the Abstract where the concept is introduced.
- **[writing]** Define ‘SFT’ and ‘RL’ at their very first occurrence in the Abstract. The text currently uses ‘supervised fine-tuning (SFT)’ and ‘reinforcement learning (RL)’ in the first sentence, but the Abstract is often read in isolation. Ensure these are explicitly defined before being used as acronyms in subsequent sentences.
- **[writing]** Replace the acronym ‘env-free’ and ‘env-based’ with ‘environment-free’ and ‘environment-based’ at their first occurrence in Section 2.1. The text defines them as ‘environment-free (env-free)’ but the hyphenated acronyms are used heavily thereafter. Consider using the full terms or ensuring the definition is prominent, as ‘env’ is informal jargon.
- **[writing]** Define ‘GRPO’ at first use in Section 2.3. The text states ‘optimize the model with GRPO’ without spelling out ‘Group Relative Policy Optimization’ or providing a citation that defines it immediately. Non-specialists may not recognize this specific RL variant acronym.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The logical consistency of the paper is generally high, with clear causal chains from the problem definition (environment setup costs) to the proposed solution (agentic verification) and the resulting benefits (scalable post-training). However, there are two areas where the premises and conclusions require tighter alignment to avoid potential logical gaps.

First, the central claim that the verifier is “environment-free” (Abstract, Sec 1) is slightly

at odds with the training methodology described in Appendix D.1. The verifier is trained via rejection sampling on trajectories where the ground-truth verdict ( $r^\star$ ) is obtained by “running the held-out unit tests” in a per-repository Docker environment. While the *inference* phase is indeed environment-free, the *training* phase fundamentally relies on environment-based execution to generate the supervision signal. The paper logically implies that the verifier learns to mimic execution without executing, but the premise that it is “environment-free” in its entirety glosses over the dependency on environment-based ground truth for training. To maintain strict logical consistency, the authors should explicitly distinguish between the “environment-free inference” capability and the “environment-dependent training data” source, ensuring the reader does not infer that the verifier learns in a vacuum.

Second, the latency analysis in Section 3.5 presents a logical leap regarding the impact on total training time. The authors correctly calculate that the verifier adds a small percentage (7.2%) to the *per-rollout* wall-clock time. They then conclude that this “does not slow down end-to-end training.” While the per-step overhead is low, the logical connection to total training time assumes that the number of training steps required for convergence remains constant regardless of the reward signal quality. If the stronger verifier leads to faster convergence (fewer steps) or if the rollout distribution changes (e.g., agents terminate earlier due to better feedback), the total time savings could be substantial. Conversely, if the stronger verifier causes the agent to explore more complex paths, the rollout time might increase. The current argument relies on the premise that rollout time is the sole bottleneck and is invariant to the verifier, which is a strong assumption not fully supported by the provided data. A more rigorous logical link would acknowledge that the *per-step* cost is negligible, but the *total* training time impact depends on convergence dynamics, which are not explicitly analyzed.

Overall, the paper’s core argument holds, but these nuances should be addressed to ensure the conclusions follow strictly from the stated premises.

#### Action items.

- **[writing]** The logical consistency of the paper is generally high, with clear causal chains from the problem definition (environment setup costs) to the proposed solution (agentic verification) and the resulting benefits (scalable post-training). However, there are two areas where the premises and conclusions require tighter alignment to avoid potential logical gaps. First, the central claim that the verifier is “environment-free” (Abstract, Sec 1) is slightly at odds with the training methodology describe

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several strong claims regarding the “fully environment-free” nature of the proposed pipeline and its performance parity with environment-based methods. While the core

contribution of using agentic exploration to replace per-repository Docker setup is novel and valuable, the current phrasing occasionally overreaches the evidence provided.

First, the term “environment-free” is used to imply a complete lack of execution or environment dependency (Abstract, Introduction). However, Section 4.1 (“Environment-free RFT”) and Appendix D.2 clarify that the agent operates in a “minimal Linux image” and that OpenHands tools “may include execution feedback from running standard developer utilities.” Furthermore, the verifier itself uses shell tools (grep, find) which are executed. The system is “Docker-free” or “setup-free,” but not strictly “execution-free.” Claiming it is “environment-free” risks misleading readers into thinking no code execution occurs at all, which contradicts the methodology where shell commands are run to gather evidence. This terminology should be tightened to reflect that the *per-repository environment construction* is eliminated, not the execution of tools entirely.

Second, the claim that the resulting model “matches the performance of standard environment-based post-training” (Abstract, Conclusion) is slightly too strong given the numerical results. Table 1 shows that on SWE-bench Multilingual, the env-free model achieves 50.0% while the env-based oracle (Test-Execution RL) achieves 51.3%. On SWE-bench Pro, the gap is 35.2% vs 35.7%. While these differences are small, “matching” suggests equivalence. The data supports that the method “approaches” or is “comparable to” the environment-based baseline, but the slight deficits on two of the three benchmarks suggest it has not fully closed the gap. The text should be adjusted to reflect this nuance to avoid overclaiming parity.

Finally, the specific claim of a “14.3 AUC points” improvement over the strongest open-source verifier (Abstract, Introduction) is derived from a custom trajectory-level benchmark described in Section 5.2 and Appendix D.3. The paper does not fully detail the distribution of this benchmark or how representative it is of the broader SWE-bench distribution. If the benchmark is biased or small, this specific margin may not generalize to all real-world scenarios. The authors should add a caveat in the limitations or discussion section acknowledging that this specific performance gain is measured on a constructed benchmark and may vary in other settings.

#### Action items.

- **[writing]** The term ‘fully environment-free’ (Abstract) overclaims. The method uses shell tools and may execute developer utilities (Sec 4.1, App D.2). It is ‘Docker-free’ or ‘setup-free’, not execution-free. Refine terminology to avoid implying zero execution occurs.
- **[writing]** Claiming the model ‘matches’ environment-based performance (Abstract) is too strong. Table 1 shows deficits on Multilingual (50.0 vs 51.3) and Pro (35.2 vs 35.7). Use ‘approaches’ or ‘comparable to’ instead of ‘matches’ to accurately reflect the data.
- **[science]** The ‘14.3 AUC’ gain (Abstract) relies on a custom benchmark (Sec 5.2) with unspecified distribution details. Add a limitation noting this margin may not generalize if the benchmark is not representative of the full SWE-bench distribution.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_safety_ethics.md`

The paper proposes “Dockerless,” an agentic verifier that explores code repositories to judge patch correctness without executing tests. From a safety and ethics perspective, the primary concern is the **dual-use potential and operational safety** of an agent system granted shell access to arbitrary codebases.

**1. Operational Safety and Sandboxing (Section 3.1, Appendix B):** The methodology explicitly states that sub-agents explore repositories using “read-only shell tools (e.g., `find`, `grep`, `rg`)” (Section 3.1). However, the prompt templates in Appendix B (e.g., **Sub-agent Per-Question Instance Template**) instruct the agent to “Execute ONE command per response” and “Chain commands with `&&` or `||`”. While the system claims to be “read-only,” the execution of arbitrary shell commands on potentially untrusted or complex file systems carries inherent risks.

- **Risk:** Without explicit mention of robust sandboxing (e.g., `seccomp` profiles, strict container isolation, or read-only mount points enforced at the kernel level), there is a non-zero risk of the agent inadvertently modifying files (e.g., via `ln -s` tricks, race conditions, or misinterpreted commands) or triggering side-channel attacks.
- **Requirement:** The authors must explicitly detail the sandboxing architecture used to guarantee the “read-only” constraint. Merely instructing the LLM to be read-only is insufficient for safety-critical infrastructure.

**2. Data Privacy and Proprietary Code (Appendix A.1):** The training data is constructed from SWE-Gym and Multi-SWE-RL, which contain real-world GitHub issues and patches.

- **Risk:** The agentic exploration process involves the model “reading” the entire repository context to answer verification questions. If the training data or the repositories used during inference contain proprietary code, secrets (API keys, passwords), or PII, the verifier could inadvertently process, memorize, or leak this information.
- **Requirement:** The paper should include a statement confirming that the training datasets were screened for sensitive information (PII, secrets) and that the inference pipeline includes mechanisms to prevent the leakage of proprietary code from the target repositories.

**3. Bias in Rejection Sampling (Section 3.2):** The verifier is trained by rejecting trajectories where the model’s verdict does not match the ground-truth test outcome.

- **Risk:** This approach assumes the ground-truth tests are unbiased. However, test suites can encode biases (e.g., favoring specific coding styles, languages, or even demographic patterns if the code is generated by specific groups). If the verifier learns to align with these biased test outcomes, it could systematically penalize valid but stylistically different solutions or specific coding patterns.
- **Requirement:** A brief discussion on the potential for bias in the training signal (derived from test suites) and any steps taken to mitigate it would strengthen the ethical rigor of the

work.

**Conclusion:** The paper presents a significant step toward scalable coding agents but introduces new safety vectors via its agentic exploration mechanism. The lack of explicit details on sandboxing and data privacy handling prevents a full “accept” verdict. Addressing these points is necessary to ensure the system can be deployed safely in real-world environments.

**Action items.**

- **[science]** The paper describes an agentic verifier that executes shell commands (find, grep, rg) on arbitrary repositories (Sec 3.1, App B). While labeled ‘read-only’, the system lacks explicit safeguards against side-channel attacks or unintended file system interactions (e.g., via symlinks or race conditions). The authors must explicitly state the sandboxing measures (e.g., seccomp, container isolation) used to prevent the verifier from modifying or exfiltrating data from the target repositories.
- **[science]** The training data construction (App A.1) relies on ‘execution-labeled’ patches from SWE-Gym and Multi-SWE-RL. The paper does not address whether these datasets contain personally identifiable information (PII), proprietary code, or sensitive credentials that might be inadvertently exposed during the verifier’s agentic exploration or included in the training corpus. A statement on data privacy and the exclusion of sensitive artifacts is required.
- **[writing]** The verifier is trained via rejection sampling on trajectories where the model’s verdict matches the ground-truth test outcome (Sec 3.2). This process risks reinforcing biases present in the test suites (e.g., favoring specific coding styles or languages) without explicit mitigation. The authors should discuss potential biases in the training signal and how they ensure the verifier does not learn to favor specific demographic or stylistic patterns in code.

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The scientific evidence supporting the central claims of Dockerless is generally robust in terms of experimental design but lacks necessary statistical rigor to fully validate the magnitude of the reported improvements.

The primary concern lies in the statistical significance of the verifier performance gains. The paper reports a 14.3 AUC point improvement (81.0 vs 66.7) on a verifier evaluation benchmark of only 776 samples (Sec 4.2, Table 2). While the absolute difference appears large, the sample size is relatively small for high-variance LLM evaluation tasks. The authors do not report confidence intervals, standard errors, or results from statistical significance tests (e.g., DeLong’s test for AUC). Without these, it is impossible to rule out that the observed gain is a result of random variance in the test set composition rather than a genuine improvement in the model’s capability.

Secondly, the RL training results rely on a very limited number of optimization steps. The authors state in Appendix D.3 that they train for only “50 RL steps” with a group size of  $G=8$ . This results in a very small effective sample size for the policy updates. Furthermore, the paper reports results from a single run (implied by the lack of “mean  $\pm$  std” notation in Table 1). In RL experiments, performance can vary significantly based on random seeds, especially with short training horizons. The claim that the environment-free pipeline “matches” the environment-based baseline (62.0% vs 62.4%) is statistically fragile without reporting the variance across multiple seeds. A difference of 0.4 points is well within the typical noise floor of such experiments.

Finally, the ablation study on the number of verification questions (Section 4.4, Figure 5) presents a curve where performance drops from 81.0 ( $K=4$ ) to 79.6 ( $K=6$ ). The authors interpret this as evidence of “redundant or noisy evidence.” However, without error bars or a statistical test, this drop could simply be noise. The conclusion that the optimal number of questions is strictly 2-4 is not fully supported by the data presented.

To strengthen the scientific evidence, the authors should: (1) report 95% confidence intervals for all AUC and resolve rate metrics; (2) run the RL training with at least 3 different random seeds and report mean  $\pm$  standard deviation; and (3) perform statistical significance tests on the verifier ablation results to confirm the observed trends are real.

#### Action items.

- **[science]** The verifier evaluation benchmark (Sec 4.2, Tab 2) uses 776 samples (500 Verified, 276 Multi-SWE). The paper claims a 14.3 AUC point gain over the strongest baseline (DeepSWE Verifier, 66.7 vs 81.0). Given the sample size, the standard error for AUC is non-negligible. The authors must report confidence intervals (e.g., 95% CI via bootstrapping) or statistical significance tests (e.g., DeLong’s test) to confirm this gain is not due to random variance on a relatively small test set.
- **[science]** The RL training uses a group size of  $G=8$  (Appx D.3) and averages  $M=2$  independent verifier passes per rollout. With only 50 RL steps total, the effective number of gradient updates is extremely low. The paper lacks an analysis of the variance in the final resolve rates across different random seeds. Without multiple seeds, the reported 62.0% resolve rate could be an outlier, and the claim of ‘matching environment-based post-training’ is statistically weak.
- **[science]** The ablation study on the number of verification questions (Fig 5) shows performance fluctuation (79.6 at  $K=6$  vs 81.0 at  $K=4$ ) but does not provide error bars or standard deviations across the 500-sample benchmark. It is unclear if the drop at  $K=6$  is statistically significant or within the noise of the evaluation metric. The conclusion that ‘additional questions introduce noise’ requires statistical backing.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The statistical analysis in the paper is generally sound in its experimental design, utilizing standard benchmarks (SWE-bench variants) and appropriate metrics (AUC for verifiers, resolve rate for agents). However, the manuscript lacks rigorous statistical validation for its key comparative claims, relying heavily on point estimates without measures of uncertainty.

First, in Section 3.2 (“Verifier evaluation”), the authors claim \approach outperforms the strongest open-source verifier by 14.3 AUC points. While the magnitude of the difference is large, the benchmark size (776 samples) is moderate. The paper does not report confidence intervals for the AUC scores or perform a statistical significance test (such as DeLong’s test for correlated ROC curves) to verify that this improvement is statistically significant and not a result of sampling variance. Given the high stakes of the claim (establishing a new state-of-the-art), providing 95% confidence intervals for all AUC values in Table 2 is necessary.

Second, the ablation study in Section 3.4 (“Effect of the number of verification questions”) presents a curve where performance peaks at K=4 and fluctuates for K=6 and K=8. The authors interpret the drop at K=6 (79.6 vs 81.0) as evidence of “redundant or noisy evidence.” However, without a statistical test (e.g., a paired t-test or bootstrap confidence intervals across the 500-sample split), it is unclear if this 1.4-point drop is statistically significant or within the noise floor of the evaluation. The conclusion that “additional questions often introduce redundant... evidence” is a strong causal claim that requires statistical backing.

Third, in Section 3.1, the paper asserts that “env-free SFT matches env-based SFT” based on resolve rates of 60.6% vs 60.0% on SWE-bench Verified. These are small absolute differences. To support the claim of “matching” performance, the authors should demonstrate statistical equivalence. This typically involves calculating confidence intervals for the resolve rates (using bootstrapping over the test instances) to show that the intervals overlap significantly or that the difference is not statistically distinguishable from zero. Currently, the text treats the point estimates as definitive proof of equivalence, which is statistically weak.

Finally, the latency analysis in Section 3.5 reports mean values (e.g., 2308s for rollout) but does not provide standard deviations or distributions for the reward evaluation times (41–180s). While the authors argue the overhead is small, the variance in reward latency could impact the stability of the RL training if not accounted for. Including standard deviations or box plots in Figure 5 would strengthen the latency argument.

In summary, while the experimental setup is robust, the statistical reporting is insufficient to fully support the claims of superiority and equivalence. The authors should add confidence intervals and significance tests to the main results and ablation studies.

#### Action items.

- **[science]** In Section 3.2 (Verifier evaluation), the paper reports AUC improvements (e.g., +14.3 points) without providing confidence intervals or statistical significance tests (e.g., DeLong’s test) to confirm these gains are not due to random variance on the 776-sample benchmark.
- **[science]** Section 3.4 (Question ablation) and Figure 4 show performance fluctuations for

K=6 and K=8. The text attributes this to ‘redundant or noisy evidence’ but lacks a statistical test (e.g., paired t-test or ANOVA) to confirm these drops are significant compared to K=4.

- **[science]** In Section 3.1, the claim that ‘env-free SFT matches env-based SFT’ relies on small absolute differences (e.g., 60.6 vs 60.0). The paper should report confidence intervals for these resolve rates (likely via bootstrapping over the benchmark instances) to demonstrate statistical equivalence rather than just point-estimate proximity.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript is generally well-written, with a clear narrative flow and professional tone. The introduction effectively sets up the problem and the proposed solution. However, there are a few specific areas where clarity and consistency need improvement before final publication.

First, there is a numerical inconsistency in Section 3.2. The text claims `\approach` improves over the strongest frontier LLM judge by “5.1 and 8.2 points, respectively” on the Verified and Multi-SWE splits. However, Table 2 shows the Multi-SWE score for GPT-5.4 is 59.5 and `\approach` is 72.1, a difference of 12.6 points, not 8.2. This discrepancy between the text and the table data must be resolved to ensure accuracy.

Second, in Section 2.2, the description of the scoring formula is accompanied by a commented-out LaTeX block containing a reviewer’s note: “`\gu{this formular is hard to understand}`”. While the formula itself is standard softmax, the presence of this internal comment in the final source is unprofessional. The authors should ensure the explanation is clear enough to stand without such notes, or remove the comment entirely.

Third, in Section 2.3, the text mentions capping the negative-to-positive sample ratio at “ $\rho$ ”. The symbol is introduced but not defined with a value in the main text. While the appendix may contain the specific value, the main text should explicitly state the value (e.g., “at  $\rho = 4$ ”) to make the methodology self-contained and reproducible.

Finally, the caption for Figure 5 (latency-overhead) in the source is incomplete or generic (“Per-rollout wall-clock breakdown...”). It should be expanded to explicitly describe the components shown in the chart (e.g., “Agent rollout time vs. reward evaluation time for three reward sources”) to match the detail provided in the main text.

Overall, the writing is strong, but these specific corrections regarding data consistency, definition of variables, and cleanup of source artifacts are necessary.

### Action items.

- **[writing]** In Section 2.2 (Architecture), the text states ‘convert the logits... into a continuous score’ followed by a softmax formula. However, the LaTeX source contains a commented-out block with a reviewer note: “`\gu{this formular is hard to understand}`”. The authors should

either clarify the formula in the main text or remove the comment to ensure the final manuscript is clean and the math is clearly explained.

- **[writing]** In Section 2.3 (Training), the text mentions ‘We additionally cap the negative-to-positive sample ratio at  $\rho$ ’. The symbol  $\rho$  is introduced but never defined with a specific value or range in the main text. While details may be in the appendix, the main text should define this hyperparameter (e.g., ‘at  $\rho = 4$ ’) to ensure the method is self-contained.
- **[writing]** In Section 3.2 (Verifier evaluation), the text states ‘improves by 5.1 and 8.2 points, respectively’ when comparing to the strongest frontier LLM judge. However, the table shows GPT-5.4 at 75.9 (Verified) and 59.5 (Multi-SWE). The calculation for Multi-SWE is  $72.1 - 59.5 = 12.6$ , not 8.2. The text claims 8.2, which contradicts the table data. This numerical inconsistency must be corrected.
- **[writing]** In Section 3.4 (Latency analysis), the text says ‘reward evaluation adds only 41–180s’. The figure caption for Fig 5 (latency-overhead) is empty in the source (‘\caption{Per-rollout wall-clock breakdown...}’). The caption should be completed to describe the figure content clearly, as the text relies on it for context.